Ryan Strawbridge

Portfolio | LinkedIn | strawbridge.r@northeastern.edu | (508) 594-9661

EDUCATION

Northeastern University Boston, MA

Candidate for B.S. in Mechanical Engineering (BSME) May 2027

Candidate for M.S. in Mechanical Engineering, Conc. in Mechanics & Design (MSME) December 2028

GPA: 3.54
Relevant Courses: Heat Transfer | Fluid Mechanics | Solid Mechanics | Thermodynamics | Material Science | Dynamics
Activities: Club Running President | Rhythm Guitarist in Alt. Rock Band
Achievements: DoD Security Clearance | CSWA Certified | Lean Six Sigma Certified

WORK EXPERIENCE

Commonwealth Fusion Systems Devens, MA

Manufacturing Engineering Co-op January 2025 – July 2026

- Designed and prototyped a hydraulic piston–driven bending machine in Siemens NX, applying bolt safety analysis and first-principles calculations to cut operation time by 75%, reduce injury risk, and improve accuracy.
- Designed, analyzed, and load-test validated an aluminum lifting frame to 3,000 lbs for easy assembly and optimized lifts. Adhered to ASME standards and received a Professional Engineer's stamp for up to 10,000 cycles of operation, increasing tooling constraints and opening use of other equipment.
- Designed an adjustable fixture using carriages, linear rails, and sheet metal — a cheap, quick solution that reduced cable cantilevering and misalignment during the wrapping process.
- Developed a generalizable equation relating bend offset, radius, and arc length through iterative bending tests and interpolation, reducing cost and enabling faster design of future cable bending dies.
- Authored detailed work instructions with technicians covering cable assembly, joint placement, orbital cutting, and welding operations in accordance with design and process specifications.

Raytheon Missiles & Defense Andover, MA

Operations Engineering Co-op January – September 2024

- Prototyped a composite fixture using SolidWorks and PCB Gerber files to shield heat-sensitive filters from 93°C while enabling reflow of adjacent components, increasing assembly yield and reducing scrap.
- Designed and tested nine solder stencil configurations that varied placement and volume, providing channels for off-gas to escape and reducing voiding, component misalignment, and rework time.
- Developed thermal reflow profiles by refining oven parameters through thermocouple data, operator input, and X-ray inspection, adhering to IPC-J-STD-001 Class 3 requirements and reducing rework time across 15 programs.
- Built an Excel VBA tool that automatically pulls and visualizes production metrics from large databases, identifying bottlenecks and accelerating throughput improvement on the rework production line.

PROJECTS

AerospaceNU Boston, MA

Vice President | Propulsion Team Member | Intro to Propulsion Lead September 2023 – Present

- Designed a machinable swirl injector — parameterizing orifice sizes, inlets, and spray angle — using HSMWorks, MATLAB, and SolidWorks Flow Simulation to generate the CAM and optimize for mass flow and pressure drop.
- Designing high-pressure flight vehicle tanks in SolidWorks and learning Abaqus FEA to evaluate critical crack length and other failure modes for structural integrity and reusability.
- Assembled the full engine stack-up, assisted with CNC milling of components, and prepared for static-fire testing by cutting graphite seals, torquing fittings, and calibrating pressure transducers.
- Redesigned a sensing PCB in KiCAD to increase the number of pressure transducers and integrate new connectors, streamlining test data collection during static fires.

SKILLS

Software: Siemens NX | SolidWorks | Ansys Mechanical | MATLAB | Excel VBA | KiCAD | Arduino | HSMWorks | Manufacturo | Odoo
Fabrication: CNC Milling | Power Tools | 3D Printing | Soldering | TIG Welding | Water Jetting | Laser Cutting
Personal: Long Distance Running | Guitar | Interior Design